

Maram Sripathi

Co2-At1

ent1

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Q1 Smart Document Editor using String, StringBuilder and StringBuffer10 marks

Question 1 :

- 1.Read a paragraph from the user;
- 2.Count the number of characters, words, and sentences.
- 3.Replace every occurrence of a given word with another word.
- 4.Reverse each sentence individually.
- 5.Maintain editing history using StringBuffer.
- 6.Construct the final formatted document using StringBuilder.
- 7.Display the original document, editing history, and final document.

Requirement: Use String methods wherever appropriate. Use StringBuilder for document construction and StringBuffer for edit history.

Your Code

```
1 import java.util.*;
2
3 public class SmartDocumentEditor
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc=new Scanner(System.in);
8
9         String paragraph=sc.nextLine();
10
11         int characters=paragraph.length();
12
13         String words[]=paragraph.trim().split("\\s+");
14         int wordCount=words.length;
15
16         String sentences[]=paragraph.split("[.!?]");
17         int sentenceCount=0;
18
19         for(String s:sentences){
20             if(!s.trim().isEmpty()){
```

Input

Java is easy. java is powerful
java
python

Output

```
Paragraph :
Editing History:
Replaced java With python
Final Document :
yesd si awaJ. lufrewop si oahryp.
```



SIMATS Online Programming Lab
 JAVA PROGRAMMING

SRIPATHI MARAM
 TRESTHARA

CO2-AT1-Assignment1
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QUESTIONS

Q1
 Smart Document Editor using
 String, StringBuffer and
 StringReader
 10 marks

Q2
 Hospital Token Management
 System using List Interface
 10 marks

Q3
 Airport Boarding Queue
 Management using Queue
 Interface
 10 marks

Q4
 Digital Certificate verification
 using JAR Interface
 10 marks

Q5
 Smart Inventory Management
 using Map Interface
 10 marks

5/5 answered

Q2 Hospital Token Management System using List Interface

10 marks

Hospital Token Management System using List Interface
 Store patient details using an appropriate List implementation.
 Register new patients.
 Display patients in registration order.
 Update patient information using Patient ID.
 Remove cancelled appointments.
 Display all patients of a particular department.
 Find the first and last registered patients.
 Requirement: Use a suitable List implementation

Your Code

```

1 import java.util.*;
2
3 class Patient
4 {
5     int id;
6     String name;
7     String department;
8
9     Patient(int id,String name,String department)
10    {
11        this.id=id;
12        this.name=name;
13        this.department=department;
14    }
15
16    public static void main(String[] args)

```

Input

```

100
Cardiology

```

Output

```

100 Rahul Cardiology
101 Anita Cardiology
102 Anil Cardiology
103 Rahul
104 Anil
105 Anil

```

Q3 Airport Boarding Queue Management using Queue Interface

10 marks

Airport Boarding Queue Management using Queue Interface
 Add passengers to the boarding queue.
 Display the current queue.
 Board passengers in FIFO order.
 Display the next passenger without removing them.
 Handle VIP passengers using a priority-based queue.
 Display remaining passengers after each boarding operation.
 Requirement: Use appropriate Queue implementations.

Your Code

```

1 import java.util.*;
2
3 class Passenger
4 {
5     String name;
6
7     Passenger(String name)
8     {
9         this.name=name;
10    }
11
12    public static void main(String[] args)
13    {
14        Scanner sc= new Scanner(System.in);
15
16        Queue<Passenger> queue = new LinkedList<>();
17        PriorityQueue<String> vip = new PriorityQueue<>();
18
19        int n=sc.nextInt();

```

Input

```

4
Rahul
Anita
Kiran

```

Output

```

Current queue>
Rahul
Anita
Kiran
Ravi
Next Passenger>

```

QUESTIONS

Q1
Smart Document Editor using
String, StringBuffer and
StringBuilder
10 marks

Q2
Hospital System Management
System using List Interface
10 marks

Q3
Airport Boarding Queue
Management using Queue
Interface
10 marks

Q4
Digital Certificate Verification
using Set Interface
10 marks

Q5
Smart Inventory Management
using Map Interface
10 marks

5/5 answered

Q4 Digital Certificate Verification using Set Interface

10 marks

Digital Certificate Verification using Set Interface

Store certificate IDs using an appropriate Set implementation.

Prevent duplicate certificate entries.

Verify whether a certificate ID exists.

Display certificates in insertion order.

Display certificates in sorted order.

Find the total number of unique certificates issued.

Remove revoked certificates.

Requirement: Use suitable Set implementations.

Your Code

```
1 import java.util.*;
2
3 public class DigitalCertificateVerification
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc=new Scanner(System.in);
8
9         LinkedHashSet<String> certificates=new LinkedHashSet<>();
10
11         int n=sc.nextInt();
12         sc.nextLine();
13
14         for(int i=0;i<n;i++)
15         {
16             certificates.add(sc.nextLine());
17         }
18
19         System.out.println("Certificates in Insertion Order:");
```

Input

C101
C104
C102
C103

Output

Certificates in Insertion Order:
C101
C104
C102
C103

Q5 Smart Inventory Management using Map Interface

10 marks

Smart Inventory Management using Map Interface

Store product information using an appropriate Map implementation.

Add new products.

Update product quantity.

Search a product using Product ID.

Calculate the total inventory value.

Display products category-wise.

Remove discontinued products.

Display all available products.

Requirement: Use a suitable Map implementation.

Your Code

```
1 import java.util.*;
2
3 class Product
4 {
5     int id;
6     String name;
7     String category;
8     int quantity;
9     double price;
10
11     Product(int id,String name,String category,int quantity,double price)
12     {
13         this.id=id;
14         this.name=name;
15         this.category=category;
16         this.quantity=quantity;
```

Input

15
101
electronics
100

Output

100 Laptop
102 Mouse
Available Product:
101 Laptop electronics 5 500000.0
102 Mouse electronics 15 100.0